

ACN TS1 QB SOLUTION – 2026-27

1. List the key roles of ICANN.

Ans.

- Manages domain names, IP address allocation, and protocol identifiers.
- Ensures uniqueness and coordination of DNS across the internet.

2. State the Subnet Address.

Ans. A **Subnet Address** is the address that identifies a particular subnet within a network.

Example:

- IP Address: 192.168.1.25
- Subnet Mask: 255.255.255.0
- Subnet Address: **192.168.1.0**

3. State the concept of fragmentation in IPv4.

Ans. Fragmentation is the process of dividing actual data into multiple fragments. Fragmentation is performed when a large packet needs to be transmitted. However, Ethernet protocol in Data Link Layer has MTU(Maximum Transfer Unit) of 1500 bytes. Therefore, when a packet is larger than 1500 bytes then it needs to be fragmented.

4. State the purpose of queuing in routers.

Ans. Queueing refers to the process of temporarily storing packets in memory buffers before they are transmitted over an output link. It is an essential function in routers because packets do not always arrive at the same rate as they can be forwarded. If too many packets arrive at once, they need to wait their turn before being sent, which is managed through queues.

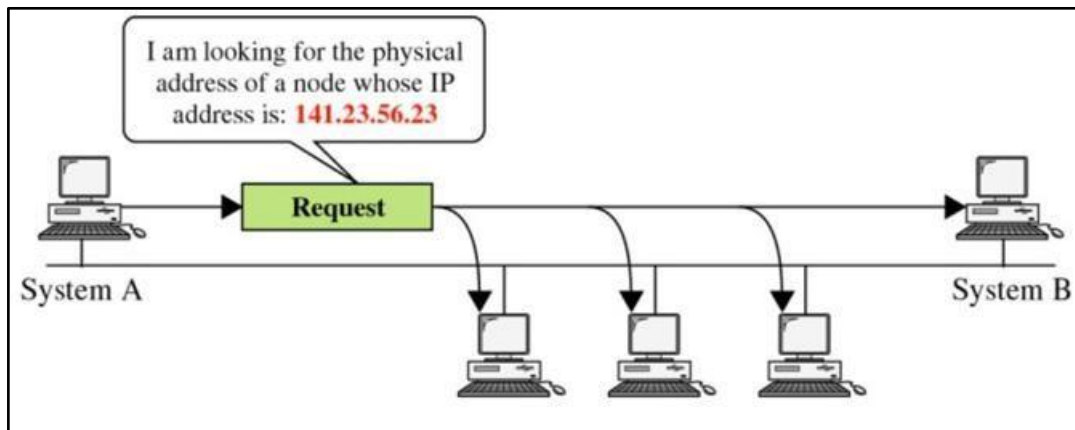
4 marks

5. Illustrate the ARP address mapping process(working) with neat diagram.

Ans. 1. ARP Query Packet Broadcast

ARP Query Packet contains

- Senders IP Address
- Senders Physical Address
- Receivers IP Address (Whom Physical Address to be found)



2. Every host or router on the network receives and examines the ARP request packet. However, only the device with the matching IP address recognizes the request and replies by sending an ARP response packet.

3. ARP Response Packet Unicast

The response packet includes the IP address and physical (MAC) address of the replying device. This packet is sent directly (unicast) to the original sender using the physical address provided in the ARP request.

Fig : ARP working (Query Packet Broadcast)

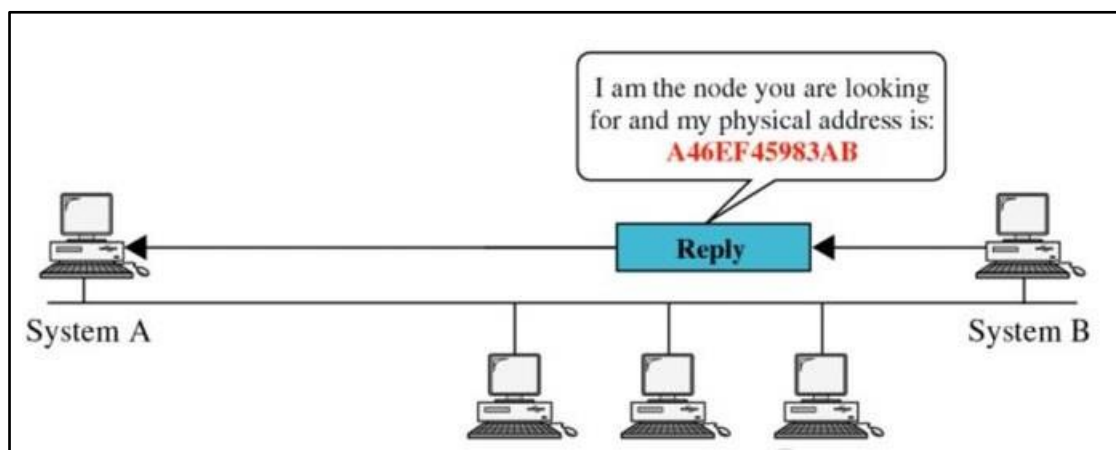


Fig : ARP working (Unicast response)

6. Explain Internet architecture with suitable diagram.

Ans.

Today an Internet has a multi-backbone structure and these backbone networks run by different private corporations today.

1. Several **backbone networks**, operated by private communication companies, provide global internet connectivity. These backbones are interconnected through peering points, enabling data exchange between them.
2. Below the backbone level, **provider networks** utilize these backbone

connections to offer internet services to end-users and organizations.

3. At the lowest level, **customer networks** access the internet through the services offered by provider networks.

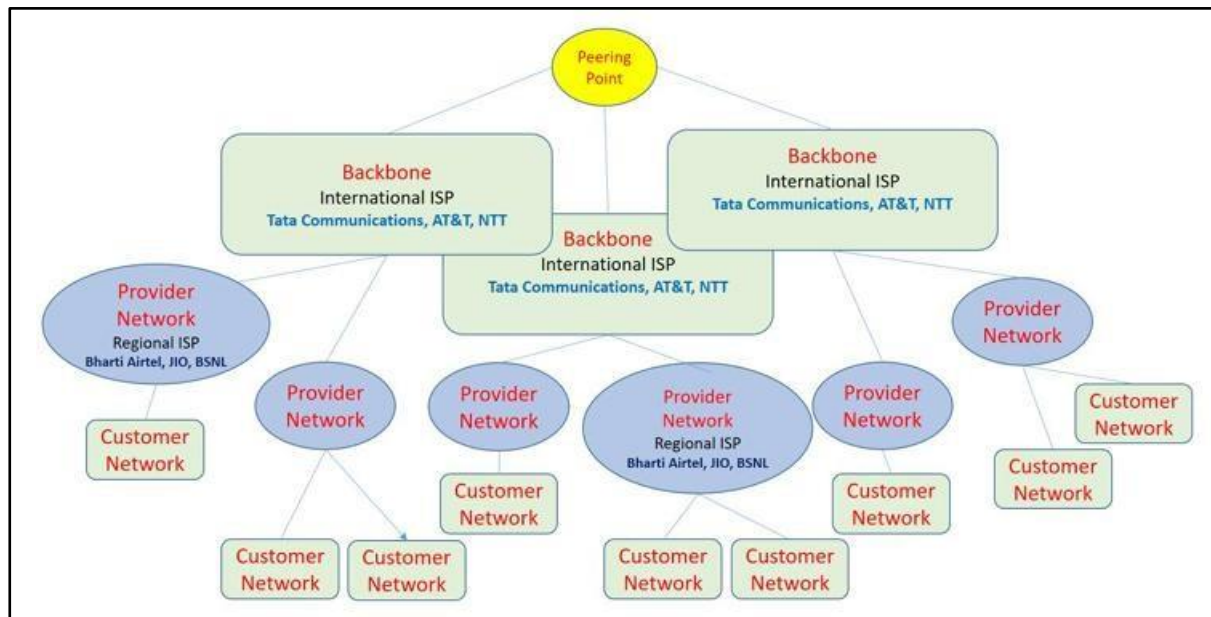


Fig : Structure of Internet

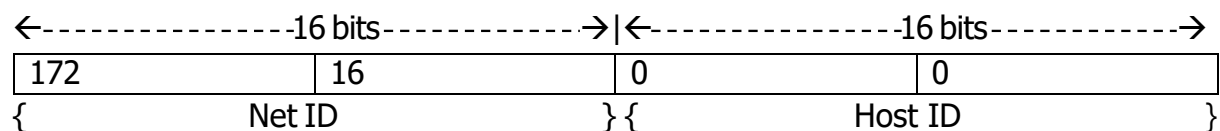
7. A college receives 172.16.0.0/16 and wants separate networks for different 8 departments. Design subnet for each department that can accommodate at least 500 hosts per department.

Identify

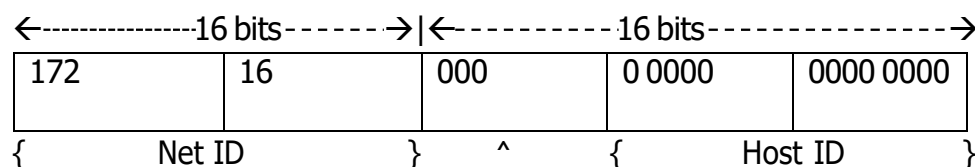
- i. Subnet Mask
- ii. Network IP
- iii. Usable Host IP range
- iv. Broadcast address

Ans.

Address 172.16.0.0/16 belongs to Class B.



To represent 8 subnets we require 3 bits ($2^3 = 8$), Borrow it from existing Host bits.



Borrowed for subnet ID

Subnet 1

172. 16. **0000** 0000. 0000 0000 → 172. 16. 0. 0
172. 16. **0000** 0000. 0000 0001 → 172. 16. 0. 1
172. 16. **0001** 1111. 1111 1110 → 172. 16. 31. 254
172. 16. **0001** 1111. 1111 1111 → 172. 16. 31. 255

Subnet mask : 255. 255. 224. 0

Network IP : 172. 16. 0. 0

First IP : 172. 16. 0. 1

Last IP : 172. 16. 31. 254

Broadcast IP : 172. 16. 31. 255

Subnet 2

172. 16. **0010** 0000. 0000 0000 → 172. 16. 32. 0
172. 16. **0010** 0000. 0000 0001 → 172. 16. 32. 1
172. 16. **0011** 1111. 1111 1110 → 172. 16. 63. 254
172. 16. **0011** 1111. 1111 1111 → 172. 16. 63. 255

Subnet mask : 255. 255. 224. 0

Network IP : 172. 16. 32. 0

First IP : 172. 16. 32. 1

Last IP : 172. 16. 63. 254

Broadcast IP : 172. 16. 63. 255

Subnet 3

172. 16. **0100** 0000. 0000 0000 → 172. 16. 64. 0
172. 16. **0100** 0000. 0000 0001 → 172. 16. 64. 1
172. 16. **0101** 1111. 1111 1110 → 172. 16. 95. 254
172. 16. **0101** 1111. 1111 1111 → 172. 16. 95. 255

Subnet mask : 255. 255. 224. 0

Network IP : 172. 16. 64. 0

First IP : 172. 16. 64. 1

Last IP : 172. 16. 95. 254

Broadcast IP : 172. 16. 95. 255

Subnet 4

172. 16. **0110** 0000. 0000 0000 → 172. 16. 96. 0
172. 16. **0110** 0000. 0000 0001 → 172. 16. 96. 1
172. 16. **0111** 1111. 1111 1110 → 172. 16. 127. 254
172. 16. **0111** 1111. 1111 1111 → 172. 16. 127. 255

Subnet mask : 255. 255. 224. 0

Network IP : 172. 16. 96. 0

First IP : 172. 16. 96. 1
Last IP : 172. 16. 127. 254
Broadcast IP : 172. 16. 127. 255

Subnet 5

172. 16. **1000** 0000. 0000 0000 → 172. 16. 128. 0
172. 16. **1000** 0000. 0000 0001 → 172. 16. 128. 1
172. 16. **1001** 1111. 1111 1110 → 172. 16. 159. 254
172. 16. **1001** 1111. 1111 1111 → 172. 16. 159. 255

Subnet mask : 255. 255. 224. 0
Network IP : 172. 16. 128. 0
First IP : 172. 16. 128. 1
Last IP : 172. 16. 159. 254
Broadcast IP : 172. 16. 159. 255

Subnet 6

172. 16. **1010** 0000. 0000 0000 → 172. 16. 160. 0
172. 16. **1010** 0000. 0000 0001 → 172. 16. 160. 1
172. 16. **1011** 1111. 1111 1110 → 172. 16. 191. 254
172. 16. **1011** 1111. 1111 1111 → 172. 16. 191. 255

Subnet mask : 255. 255. 224. 0
Network IP : 172. 16. 160. 0
First IP : 172. 16. 160. 1
Last IP : 172. 16. 191. 254
Broadcast IP : 172. 16. 191. 255

Subnet 7

172. 16. **1100** 0000. 0000 0000 → 172. 16. 192. 0
172. 16. **1100** 0000. 0000 0001 → 172. 16. 192. 1
172. 16. **1101** 1111. 1111 1110 → 172. 16. 223. 254
172. 16. **1101** 1111. 1111 1111 → 172. 16. 223. 255

Subnet mask : 255. 255. 224. 0
Network IP : 172. 16. 192. 0
First IP : 172. 16. 192. 1
Last IP : 172. 16. 223. 254
Broadcast IP : 172. 16. 223. 255

Subnet 8

172. 16. **1110** 0000. 0000 0000 → 172. 16. 224. 0
172. 16. **1110** 0000. 0000 0001 → 172. 16. 224. 1
172. 16. **1111** 1111. 1111 1110 → 172. 16. 255. 254
172. 16. **1111** 1111. 1111 1111 → 172. 16. 255. 255

Subnet mask : 255. 255. 224. 0

Network IP : 172. 16. 224. 0

First IP : 172. 16. 224. 1

Last IP : 172. 16. 255. 254

Broadcast IP : 172. 16. 255. 255

8. Differentiate between IPv4 and IPv6. (Any 6 points).

Ans.

Features	IPv4	IPv6
Address Size	32 bit	128 bit
Header Size	Variable : 20-60 bytes	Fixed : 40 bytes base header
Header fields	More fields, including Header Checksum	Simplified Header; No Header Checksum
Fragmentation	Can be done by Source and routers	Done only by the source using Fragment Extension Header
Options	Included within the main header	Handled using Extension Headers
TTL / Hop Limit	Uses TTL (Time to Live)	Uses hop limit
Protocol field	Uses protocol field	Uses next header field

9. An organization has been assigned 192.168.10.0/24 and has 4 departments requiring 100, 50, 25, and 10 hosts. Design a suitable VLSM addressing scheme.

Ans.

Arrange in descending order 100, 50, 25, 10

100 Hosts

$$2^7 = 128$$

Minimum 7 bits are required for hosts addressing, therefore subnet mask will be 192. 168. 10. 0/25

Network ID : 192. 168. 10. 0

First Valid IP : 192. 168. 10. 1

Last Valid IP : 192. 168. 10. 126

Broadcast IP : 192. 168. 10. 127

50 Hosts

$$2^6 = 64$$

Minimum 6 bits are required for hosts addressing, therefore subnet mask will be 192. 168. 10. 0/26

Network ID : 192. 168. 10. 128

First Valid IP : 192. 168. 10. 129

Last Valid IP : 192. 168. 10. 190

Broadcast IP : 192. 168. 10. 191

25 Hosts

$$2^5 = 32$$

Minimum 5 bits are required for hosts addressing, therefore subnet mask will be 192. 168. 10. 0/27

Network ID : 192. 168. 10. 192

First Valid IP : 192. 168. 10. 193

Last Valid IP : 192. 168. 10. 222

Broadcast IP : 192. 168. 10. 223

10 Hosts

$$2^4 = 16$$

Minimum 4 bits are required for hosts addressing, therefore subnet mask will be 192. 168. 10. 0/28 (total 16 Ips)

Network ID : 192. 168. 10. 224

First Valid IP : 192. 168. 10. 225

Last valid IP : 192. 168. 10. 239

Broadcast IP : 192. 168. 10. 240

10. A router receives an IPv4 datagram of total size 4000 bytes.

The datagram header is 20 bytes and the outgoing network has an MTU of 1500 bytes.

- i. Why does the router need to fragment the datagram? (1 Mark)
- ii. Calculate the size of the data carried in each fragment. (2 Marks)
- iii. Determine the number of fragments generated. (1 Mark)
- iv. Explain the role of the Identification, Flags and Fragment Offset fields in reassembling these fragments at the destination. (2 Marks)

Ans.

- i. The datagram size is **4000 bytes**, which is greater than the outgoing network's **MTU of 1500 bytes**. Therefore, the router must **fragment the datagram** into smaller packets.

ii. Total datagram size = 4000 bytes

Header size = 20 bytes

MTU = 1500 bytes

Original data size : $4000 - 20 = 3980$ bytes

Maximum data per fragment : $1500 - 20 = 1480$ bytes

Therefore:

- Fragment 1 data = **1480 bytes**
- Fragment 2 data = **1480 bytes**
- Fragment 3 data = **1020 bytes**

iii. Fragment 1 = 1480

Fragment 2 = 1480

Fragment 3 = 1020

$3980 = 1480 + 1480 + 1020$

Therefore, **3 fragments** are generated.

iv. Role of Identification, Flags and Fragment Offset :-

1. **Identification** : All fragments of the same original datagram have the **same Identification value**. It helps the destination identify which fragments belong to the same datagram.
2. **Flags** :
 - First flag bit is reserved (not in use).
 - Second flag bit is Do Not Fragment. If it is 1, it instructs routers not to fragment the packet.
 - Third flag bit is, **MF (More Fragments)** flag indicates whether more fragments follow. It is **1** for the first two fragments and **0** for the last fragment.
3. **Fragment Offset** : Indicates the **position of each fragment's data** within the original datagram. The destination uses it to put the fragments back in the correct order.

Thus, the destination uses **Identification + Flags + Fragment Offset** to correctly **reassemble the original IPv4 datagram**.

Chapter 2

2 marks

11. Define:

i. Inter-domain routing

ii. Intra domain routing

Ans.

i. **Inter - domain routing** : Routing between autonomous systems is referred to as inter domain routing.

ii. **Intra - domain routing** : Routing inside an autonomous system is referred to as intra-domain routing.

12. List Intra-domain unicast and multicast protocol.

Ans. (*Write any 2*)

Based on Distance Vector Routing : -

Unicast : RIP (Routing Information Protocol)

Multicast : DVMRP

Based on Link State Routing : -

Unicast : OSPF (Open Shortest Path First)

Multicast : MOSPF

13. What is the purpose of ICMP? List two types of ICMP messages.

Ans. ICMP (Internet Control Message Protocol) is used by network devices to send error messages and diagnostic information about IP packet delivery.

Two Types of ICMP Messages :

1. **Echo Request / Echo Reply** - Used by the ping command to check whether a host is reachable.
2. **Destination Unreachable** - Sent when a packet cannot reach its intended destination.

4 marks

14. Explain distance vector routing with Example.

Ans.

- In distance vector routing, the least-cost route between any two nodes is the route with minimum distance.
- In this algorithm, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).

1. Steps to build Routing Table for each node :-

- i. Initialization
- ii. Sharing
- iii. Updating

Now consider following Autonomous System as an example with 5 nodes (Routers).

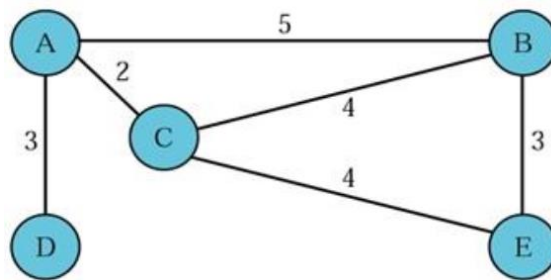
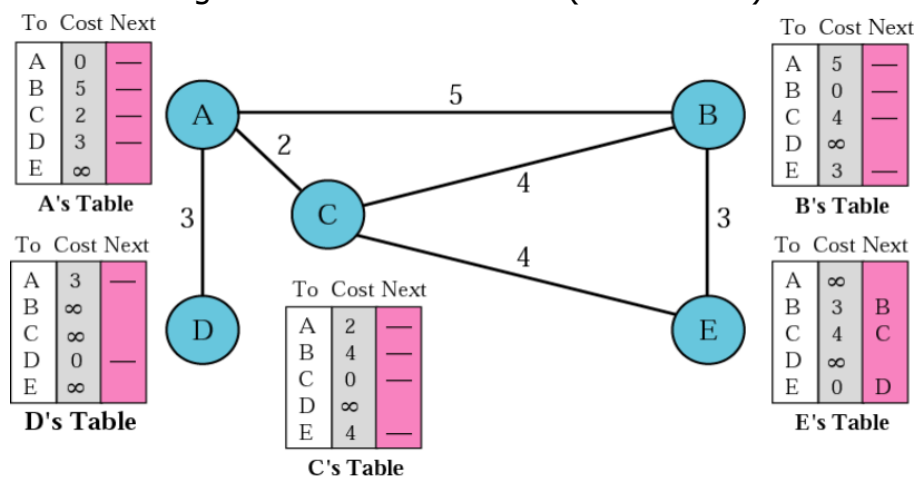


Fig. 3.1 Network

Initialization

In this step, the router can send a message to its immediate neighbors i.e. those directly connected to it and find the distance between them. The distance for any entry that is not a neighbor is marked as infinite (unreachable).

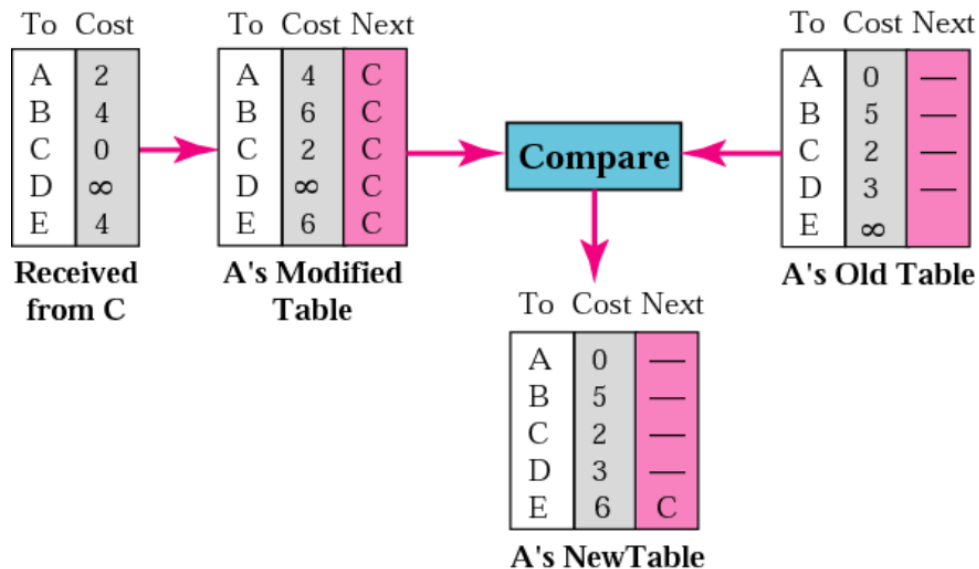


Sharing

In this step, each router shares its routing table with its immediate neighbours periodically and when there is a change.

Updating

In this step, each router takes the received routing table from other routers and updates its own existing routing table according to the other routing tables.



15. State the characteristics of the Routing Information Protocol (RIP).

(Any four points)

Ans.

- RIP protocol is based on Distance vector routing algorithm.
- RIP is an Intra-domain routing protocol.
- Maximum Hop count limit is 15, if reached to 16 it will be considered as unreachable.
- RIP suitable for small networks due to limited maximum hop count.
- RIP uses UDP service at transport layer.
- Each router sends the routing update out of all active interfaces every 30 seconds.
- Each router also receives routing updates from its neighbouring routers.

16. Describe the RIP message format along with purpose of each field.

Ans.

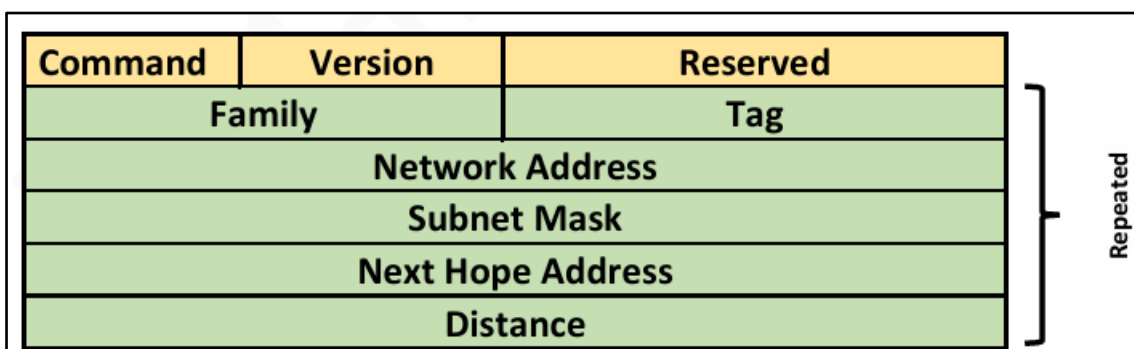


Fig : RIP Message Format

Command (8 bit) - type of message: request (1) or response (2).

Version (8 bit) - current Version 2

Family (16 bit) - Family of protocol, for TCP/IP value is 2

Tag (16 bit) - Information about autonomous system

Network Address (32 bit) - Destination address

Subnet Mask (32bit) – To identify prefix bits

Next Hope Address (32bit) – Defines via which interface packet can reach destination

Distance (32bit) - Number of hops to the destination

17. Describe Features of OSPF. (Any four)

Ans.

- It is an interior gateway protocol that has been designed to route packets within a single autonomous system.
- It is based on a link-state routing algorithm in which each router or node contains the complete topology of autonomous system, and based on this information, it form the shortest path tree.
- Every router contains the same information about the network. The way the router learns this information by sending LSA (Link State Advertisements).
- Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as LSDB.
- Divides large networks into smaller segments called areas (with Backbone Area 0 at the core) to reduce routing traffic and memory overhead.

18. Explain four functions of Border Gateway Protocol.

Ans.

- It is an Exterior gateway protocol that has been designed for routing between multiple autonomous system (Inter-domain Routing).
- It is based on a path vector routing algorithm.
- It runs over TCP port number 179, support reliability.
- Separate TCP Connection is established between BGP Speaker routers to exchange information regarding reachability to different AS.

19. Compare OSPF and RIP. (Any four points)

Justify OSPF's Effectiveness in large network.

Ans.

Feature	RIP	OSPF
Algorithm	Distance Vector	Link State
Metric	Hop count	Cost (based mainly on bandwidth)
Maximum hops	15 hops	No 15-hop limitation
Convergence	Slower	Faster
Network size	Suitable for small networks	Suitable for medium/large networks
Update frequency	Typically every 30 seconds	Sends LSAs when topology changes, with periodic refreshes
Administrative Distance	120	110
Hierarchical design	No	Yes, using areas

OSPF Effectiveness in Large Networks :- (Write any 2)

1. **Scalable:** Handles large networks efficiently.

2. **Fast Convergence:** Quickly adapts to network changes.
3. **Areas:** Divides networks into smaller areas to reduce traffic.
4. **Shortest Path:** Uses Dijkstra's algorithm to find the best path.
5. **Less Overhead:** Sends updates mainly when changes occur.
6. **Load Balancing:** Supports multiple equal-cost paths.

20. Distinguish between dynamic routing and static routing on the Basis of configuration, security, cost and routing protocols.

Ans.

Basis	Static Routing	Dynamic Routing
Configuration	Manually configured by the network administrator	Automatically updated by routing protocols
Security	More secure because routers are not shared automatically	Less secure as routing information is exchanged between routers
Cost	Low cost; requires less router resources	Higher cost due to CPU, memory and bandwidth usage
Routing Protocols	Does not use routing protocols	Uses protocols like OSPF, RIP, BGP

6 marks

21. Differentiate between

Distance vector routing and link state routing. (any 6 points)

Ans.

Distance Vector Routing	Link State Routing
1. Shares routing information with neighbouring routers.	Shares link-state information with all routers in the network/area.
2. Uses distance and direction to find routes.	Uses a complete network topology to find routes.
3. Uses algorithms such as Bellman-Ford .	Uses Dijkstra's Shortest Path First algorithm.
4. Has slower convergence .	Has faster convergence .
5. Requires less CPU and memory .	Requires more CPU and memory .
6. Example: RIP .	Example: OSPF .

22. A network administrator wants to check whether a host with IPv4 address 192.168.1.20 is reachable from a host with IPv4 address 192.168.1.10. The administrator uses the ping command.

- i. Which ICMP query message is generated by the source host? (1 Mark)
- ii. Which ICMP query message is returned by the destination host? (1 Mark)
- iii. Explain the purpose of the Identifier and Sequence Number fields in the ICMP message. (2 Marks)
- iv. If the source receives an ICMP Echo Reply, what can the administrator conclude? What cannot be concluded? (2 Marks)

Ans.

i. ICMP Echo Request with type = 8, is sent by the source host

ii. ICMP Echo Reply with type = 0, is returned by the destination host.

iii. Purpose of Identifier and Sequence Number : -

- **Identifier:** Identifies the ping session/request so that the reply can be matched with the correct request.
- **Sequence Number:** Numbers each Echo Request and Reply, helping identify the order of messages and detect missing replies.

iv. Conclusion from ICMP Echo Reply : -

Can conclude :

- The destination host is **reachable**.
- IP connectivity between the two hosts is working.
- The destination responded to the ICMP Echo Request.

Cannot conclude :

- It does **not guarantee** that all network services or applications on the destination are working.
- It does **not guarantee** that the network is completely error-free or that every packet will reach the destination.

23. A router R1 has following route in its routing table.

Destination Network	Next Hope	Metric
192.168.1.0	Directly Connected	0
192.168.2.0	R1	1
192.168.3.0	R2	2

Design “RIP Response message” for R1 to send its neighbouring router R4.

Ans.

Given routing table for R1 is

Destination Network	Next Hope	Metric
192.168.1.0	Directly Connected	0
192.168.2.0	R1	1
192.168.3.0	R2	2

RIPv2 Message Format

Command	Version	Reserved
Family		Tag
Network Address		
Subnet Mask		
Next Hope Address		
Distance		

Fig. 2.6 RIP Packet format version2

To sends its routing information to neighboring router R4, R1 will prepare a RIP Response msg. which will include all its routing records. RIP response message from R1 will looks like as follows

- Since we are preparing Response message – command will contain **2** as its value.
- RIP version is also **2**.
- Family ID for IPv4 protocol is **2**.
- Tag will indicate AS number, we will assume it as **101**.

2	2	0
2	101	
192.168.1.0		
255.255.255.0		
0		
0		
2	101	
192.168.2.0		
255.255.255.0		
R1		
1		
2	101	
192.168.3.0		
255.255.255.0		
R2		
2		

RIP Response message from R1 to R4

NOTE:

Entries in RIP message format must be as per Routing table of Sender Router in our example R1.